Ex.No.-3.2-Create 3D model from orthographic

AIM:

To create a 3D model from given orthographic projection.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

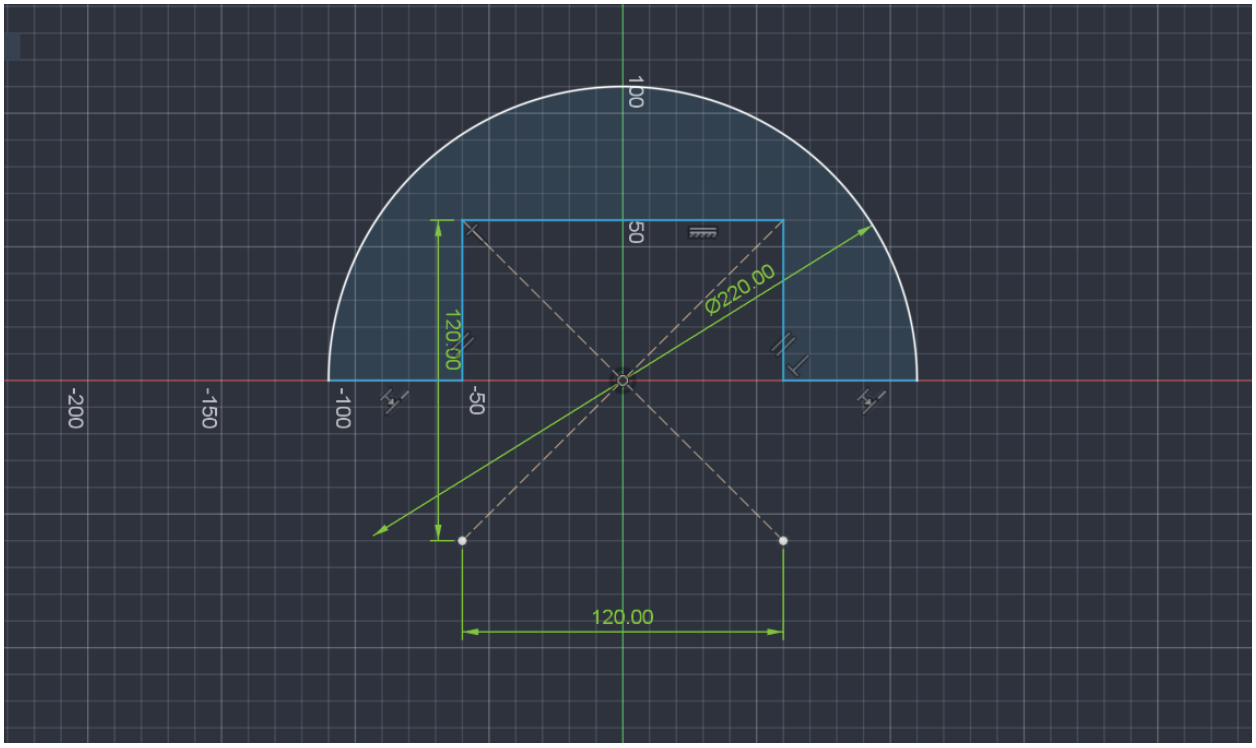
STEP-1:
Open new file and then select sketch icon.
Select the front view for sketching.

STEP-2:
Create a Sketch
Create a circle of diameter 220mm.

STEP-3:
Rectangle > Center Rectangle.
Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.
Click once to begin placing the rectangle.
Type the Tab button and enter the values 120 mm for both the width and height respectively.

STEP-4:
Draw a line at the x axis.

Trim the bottom half.



Click Finish Sketch after completing.

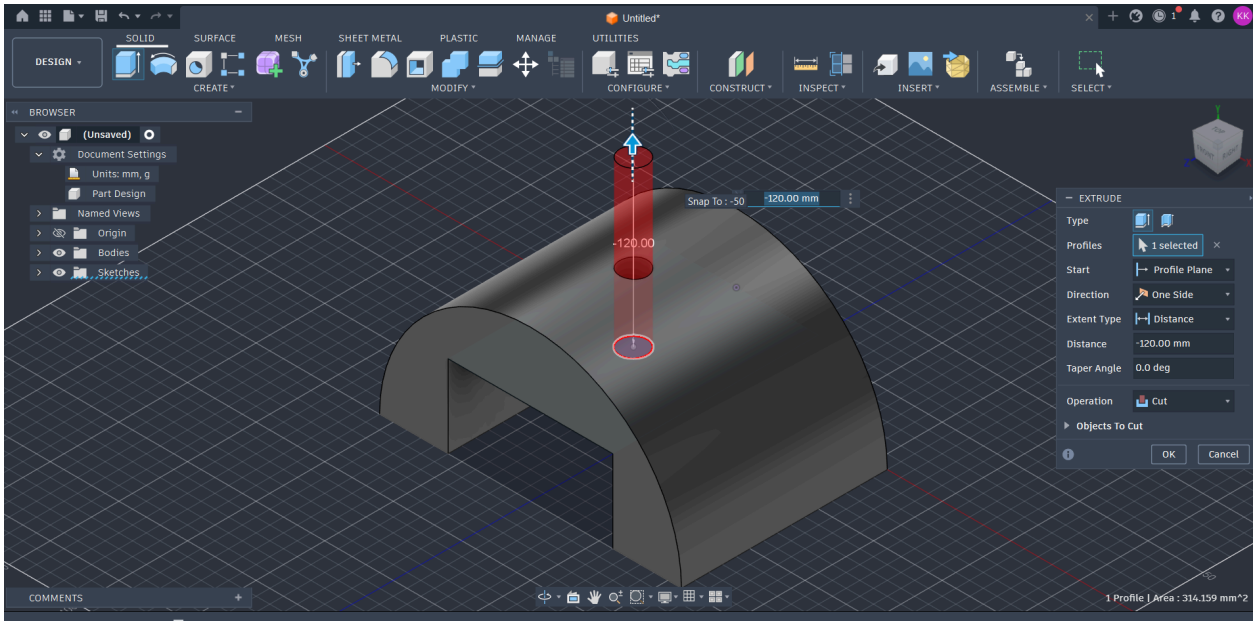
STEP-5:
Select Extrude.
Select the sketch.
Click again to place the value.
Type in the value of 150 mm.
Press ENTER to accept the value.
STEP-6:
Start the sketch command
Select Sketch > Create Sketch.

STEP-7:
Select the Sketch Plane
Select the TOP SURFACE of the 3D block

STEP-8:
Create a circle of diameter 20mm at the centre.
Click Finish Sketch after completing.

STEP-9:
Select Extrude.
Select the sketch.
Choose the cut operation.

Drag the arrow till the portion is cut out.

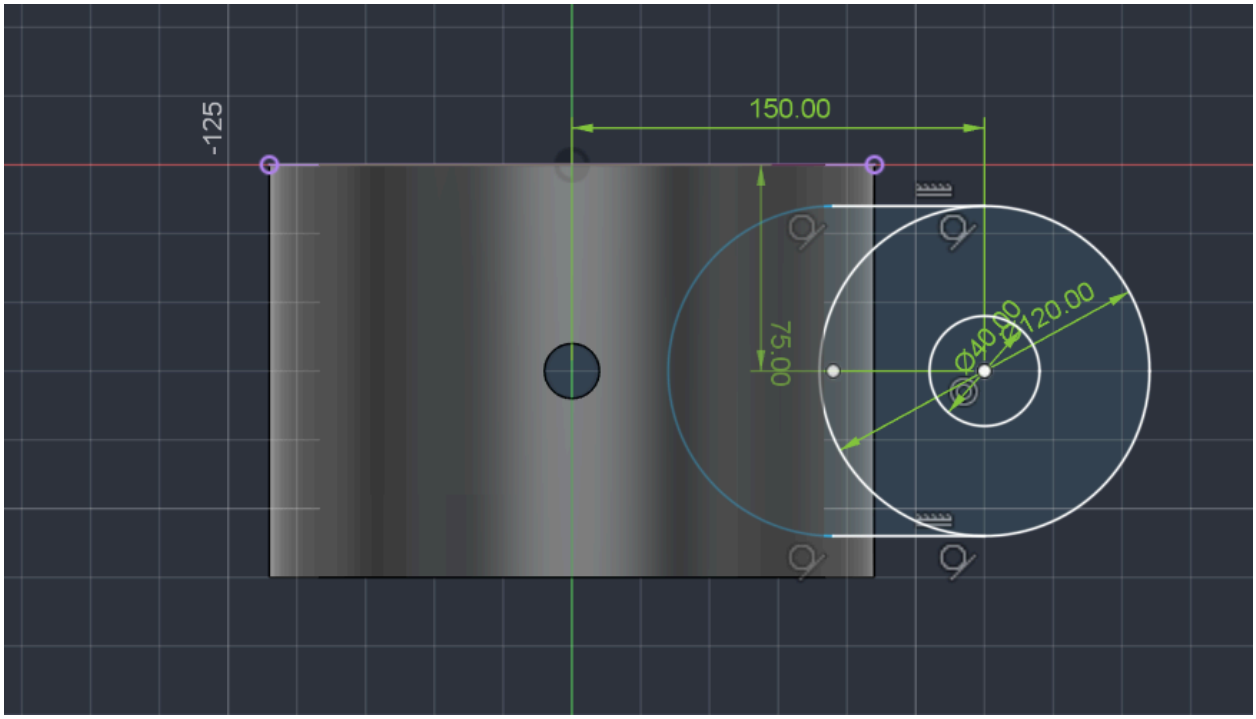


STEP-10:
Select the Sketch Plane
Select the TOP SURFACE of the 3D block

STEP-11:
Draw a circle of diameter 120mm.
The circle is to be at a distance of 150mm to the right side of the origin.
Draw a smaller circle of diameter 40mm from the same centre point of previous circle.

STEP-12:
Connect the circles and existing shape using two tangent lines.

Trim the excess.

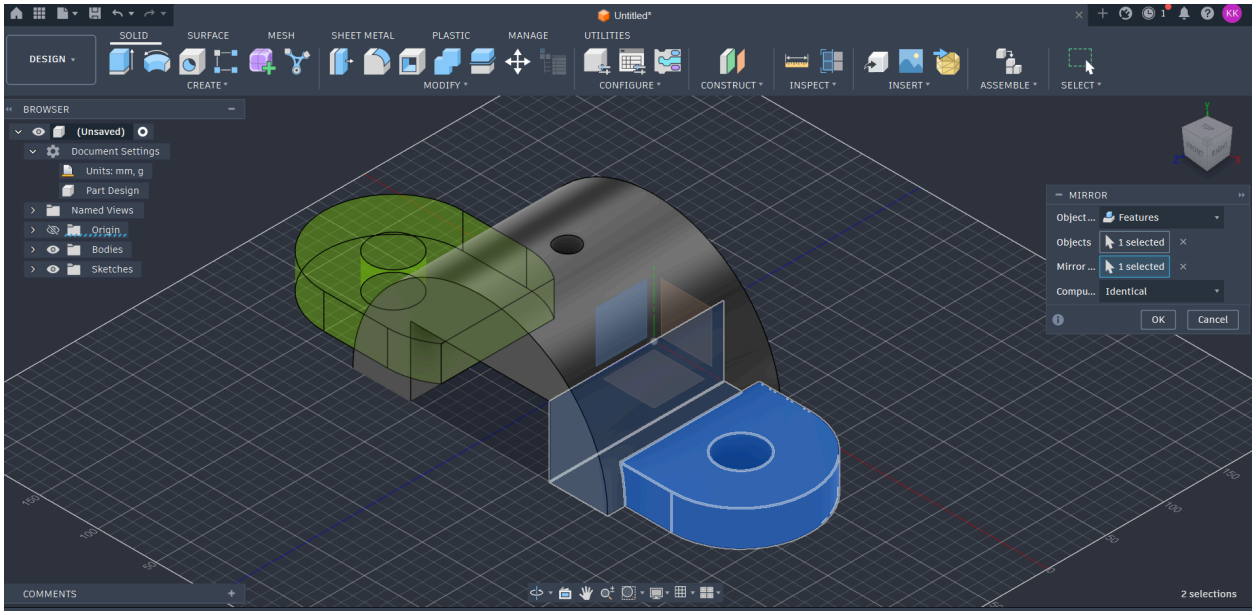


Click Finish Sketch after completing.

STEP-13:
Select Extrude.
Select the sketch.
Click again to place the value.
Type in the value of -30 mm.
Press ENTER to accept the value.

STEP-14:
Select mirror.

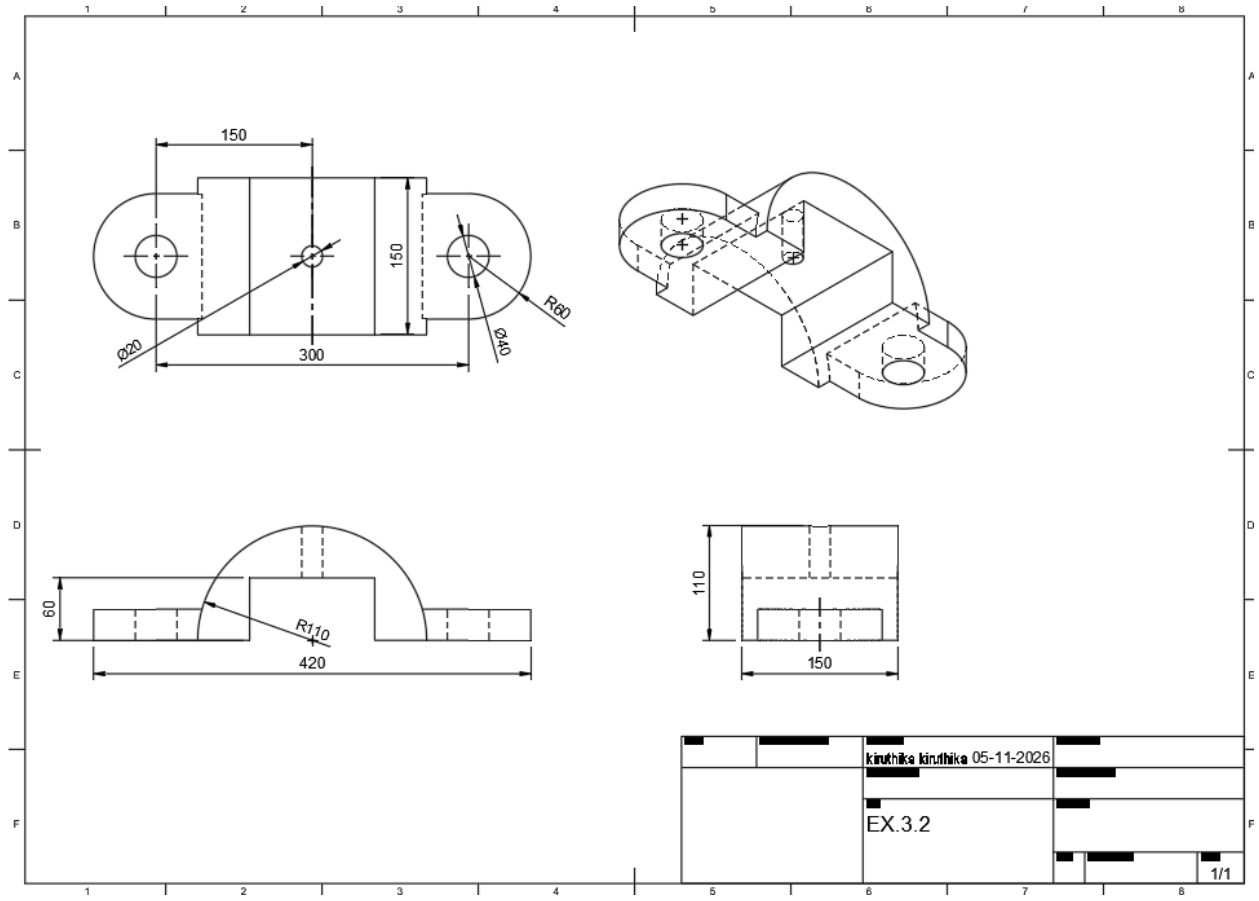
Copy the right side to the left about the vertical plane.



STEP-15:

Switch to drawing mode to plot the sketch in ISO standard.

OUTPUT:



RESULT:

Thus, 3D model is created from orthographic projection.